$D = \{(x^{(i)}, y^{(i)})\}_{i=1}^{n}$ ⇒ tree $\hat{f}(x) = \sum_{m=1}^{M} O_m \mathbb{1}[x \in R_m]$ 공간분할

Bagging / RF $\hat{f}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}_b(x)$ 평균 · 투표

NN $h(x) = y^{(i^*)}$, $i^* = \arg\min_i d(x, x^{(i)})$ 거리·유사도 측정해 거리로 예측

non-parametric method : 고정된 파라미터 개수 존재이 아니라 학습 데이터가 모델 구조를 결정한다.

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1. Decision Tree

- $\hat{f}_T(x) = \sum_{m=1}^{M} O_m \mathbb{1}[x \in R_m]$ with 공간의 분할 $R_1, \dots, R_M$
- CART/ID3 : 축정렬 분할 $x_j < s$ + 각 leaf에는 상수 O_m $x_j \geq s$ (Greedy)

회귀 트리

- $O_m = \frac{1}{N_m} \sum_{i: x^{(i)} \in R_m} y^{(i)}$ ⇒ minimize $\left[\lambda M + \sum_{m=1}^{M} E_m \right]$ is NP-Complete.
- $E_m = \sum_{i: x^{(i)} \in R_m} (y^{(i)} - O_m)^2$
- 탐욕 분할

$R_{j,s}^{-}(D) = \{(x,y) \in D : x_j < s\}$

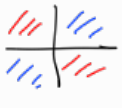
$R_{j,s}^{+}(D) = \{(x,y) \in D : x_j \geq s\}$

$(j^*, s^*) = \underset{j,s}{\arg\min} \left[E_{R_{j,s}^{-}(D)} + E_{R_{j,s}^{+}(D)} \right]$

⇒ 재귀적으로 until $|D| \leq K$

- # of 분할 = $O(dn)$
- 분할 평가 비용 = $O(n)$

Pruning : 트리를 크게 만든 뒤 줄이는 전략



XOR

- $C_\alpha(T) = \sum_{m=1}^{|T|} E_m(T) + \alpha|T|$ ↳ cross validation
- weakest link pruning : error 증가가 가장 작은 split 제거 ⇒ tree sequence에서 $C_\alpha(T)$가 가장 작은 tree

2. Classification Tree

- O_m = majority $i : x^{(i)} \in R_m$ $y^{(i)}$
- $\bar{E}_m = |\{i : x^{(i)} \in R_m,\ y^{(i)} \neq O_m\}|$
- $\hat{P}_{mk} = \dfrac{|\{i : x^{(i)} \in R_m,\ y^{(i)} = k\}|}{N_m}$
- $\hat{P}_{mj\nu} = \dfrac{|\{i : x^{(i)} \in R_m,\ x_j^{(i)} \geq \nu\}|}{N_m}$

Impurity

(i) Misclass : $Q_m = \dfrac{\bar{E}_m}{N_m} = 1 - \hat{P}_{mO_m}$ = $\min(p, 1-p)$ • 1번분류

(ii) Gini : $Q_m = \sum_k \hat{P}_{mk}(1 - \hat{P}_{mk})$ = $2p(1-p)$

(iii) Entropy : $H(R_m) = -\sum_k \hat{P}_{mk} \log \hat{P}_{mk}$ = $-p\log p - (1-p)\log(1-p)$

- Entropy를 쓰면 split 진행은 information gain 최대.

Info Gain $(X_j = \nu, R_m)$

$= H(R_m) - \left(\hat{P}_{mj\nu} H(R_{j,\nu}^{+}) + (1 - \hat{P}_{mj\nu}) H(R_{j,\nu}^{-})\right)$

- tree 성장에는 entropy,

  pruning에는 misclassification error

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3. Bagging : bootstrap 후 평균 · 투표 .

- Bootstrap Sample : $D_b \sim$ sample with replacement from D

  $|D_b| = n$

- $\hat{f}_b = T_{train}(D_b)$

- 최종 bagging : $\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)$

- 분류 $\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x)$

  $\hat{y}_{bag}(x) = \arg\max_k \hat{f}_{bag}(x)_k$

- nonlinear / adaptive

  predictor의 estimation error ↓

- 단일 tree의 해석 가능성 ↓.

Random Forest. $T_b = Tree(Bootstrap(D),$ each split : choose best among (m) random features from (d))

$\hat{y}(x) = vote\{T_b(x)\}_{b=1}^{B}$

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4. Nearest Neighbour

- $d(x,x) = 0$ $d(x,x') = d(x',x)$ $d(x,x'') \le d(x,x') + d(x',x'')$

- 1-nearest neighbor

  $i^*(x) = \arg\min_i d(x, x^{(i)})$, $h(x) = y^{(i^*(x))}$

  Voronoi Cell $V_i = \{x : d(x, x^{(i)}) \le d(x, x^{(l)}) \ \forall l\}$

- k-NN $N_k(x) =$ k nearest points

  $\hat{y}_{reg}(x) = \frac{1}{k}\sum_{i \in N_k(x)} y^{(i)}$

  $\hat{y}_{class}(x) = \arg\max_c \sum_{i \in N_k(x)} \mathbb{1}[y^{(i)} = c]$

- locally weighted regression $\hat{\theta}(x) = \arg\min_\theta \sum_{i \in N_k(x)} w_i(x)\left(y^{(i)} - \theta^T \phi(x^{(i)})\right)^2$

  $w_i(x)$ → $d(x, x^{(i)})$에 반비례.

- ball tree (대규모 데이터)